

7.2.6 Software Review Process

NOTE — Software reviews are held at both **project management** and **technical levels** throughout the software life cycle. Their purpose is to maintain a common understanding among stakeholders of project status, risks, and objectives, and to determine actions necessary to ensure that the software product satisfies stakeholder expectations.

7.2.6.1 Purpose

The purpose of the Software Review Process is to **maintain a common understanding with the stakeholders of the progress against the objectives of the agreement** and to determine what should be done to help ensure development of a product that satisfies the stakeholders. Software reviews provide structured opportunities for early detection of issues and collaborative decision-making.

7.2.6.2 Outcomes

As a result of successful implementation of the Software Review Process:

- a) management and technical reviews are held based on the needs of the project;
 - b) the status and products of an activity or process are evaluated through review activities;
 - c) review results are made known to all affected parties;
 - d) action items resulting from reviews are tracked to closure; and
 - e) risks and problems are identified and recorded.
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7.2.6.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Review Process.

7.2.6.3.1 Process Implementation

This activity consists of the following tasks:

7.2.6.3.1.1 Review Scheduling

Periodic reviews shall be held at **predetermined milestones** as specified in the project plan(s). Stakeholders may also convene **ad hoc reviews** as necessary.

7.2.6.3.1.2 Resource Provisioning

All resources required to conduct reviews shall be provided. These resources include personnel, locations, facilities, hardware, software, and tools needed to support effective review sessions.

7.2.6.3.1.3 Review Preparation

The participating parties should agree on the following for each review:

- meeting agenda;
 - software products (work products/results) and problems to be reviewed;
 - scope and procedures;
 - **entry and exit criteria** for the review.
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7.2.6.3.1.4 Problem Recording

Problems detected during reviews shall be **recorded and entered into the Software Problem Resolution Process (7.2.8)** for tracking and resolution.

7.2.6.3.1.5 Review Results Documentation

The results of the review shall be documented and distributed to all relevant parties. This includes formal review outcomes such as **approval, disapproval, or contingent approval**.

7.2.6.3.1.6 Action Tracking and Closure

Participating parties shall agree on the **outcome of the review**, assign action item responsibilities, and define **closure criteria**. Action items shall be tracked to completion.

7.2.6.3.2 Project Management Reviews

This activity consists of the following task:

7.2.6.3.2.1 Project Status Evaluation

Project status shall be evaluated relative to the applicable project plans, schedules, standards, and guidelines. The outcome of the review should be considered by appropriate management and may include:

- a) ensuring activities progress according to plan, based on an evaluation of activity or product status;
 - b) maintaining overall control of the project through adequate allocation of resources;
 - c) changing project direction or determining the need for alternate planning;
 - d) evaluating and managing risk issues that may jeopardize the success of the project.
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7.2.6.3.3 Technical Reviews

This activity consists of the following task:

7.2.6.3.3.1 Product and Process Evaluation

Technical reviews shall be held to evaluate the software products or services under consideration and provide evidence that:

- a) they are **complete**;
 - b) they comply with their **standards and specifications**;
 - c) changes are properly implemented and affect only those areas identified by the **Software Configuration Management Process (7.2.2)**;
 - d) they adhere to applicable schedules;
 - e) they are **ready for the next planned activity**; and
 - f) the development, operation, or maintenance is being conducted according to the plans, schedules, standards, and guidelines of the project.
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7.2.6.4 Deliverables

The following deliverables shall be produced during this process:

- Review schedule and agenda;
- Review checklists and criteria;

- Meeting minutes and attendance records;
 - Problem and action item logs;
 - Review results documentation (approval, disapproval, contingent approval);
 - Risk and issue records identified during reviews.
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7.2.6.5 Audit and Configuration Control

All review records, minutes, action item logs, and review reports shall be placed under **configuration management**.

Audit evidence shall be retained according to organizational and regulatory policies. Review artifacts shall be versioned and linked to corresponding baselines and work products to maintain traceability between reviews and project milestones.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — reviewed to ensure correctness and stakeholder alignment.
- **Software Architectural Design (7.1.3)** and **Detailed Design (7.1.4)** — reviewed at design milestones for completeness and compliance.
- **Software Construction (7.1.5), Integration (7.1.6), and Qualification Testing (7.1.7)** — reviewed to assess development and integration progress.
- **Software Configuration Management (7.2.2)** — provides baseline references and ensures controlled change review.
- **Software Quality Assurance (7.2.3)** — coordinates reviews to ensure quality oversight.
- **Software Verification (7.2.4)** and **Validation (7.2.5)** — provide evidence and work products subject to review.
- **Software Audit (7.2.7)** — may follow reviews to independently assess compliance.
- **Software Problem Resolution (7.2.8)** — receives issues and action items generated by reviews.